FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
FULL TEST – X
PAPER –1
TEST DATE: 15-05-2024

ANSWERS, HINTS & SOLUTIONS

Physics

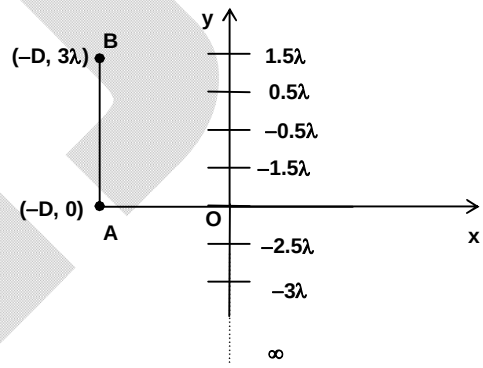
PART – I

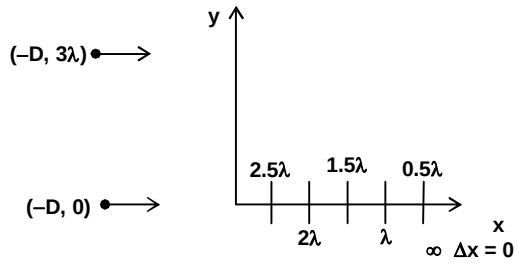
SECTION – A

1. A, B, C

Sol. Δx and ∞ on y-axis is 3λ and -3λ

There will be 6 minima and 5 maxima





2. A, D

Sol. Given $\mu = \sqrt{1+e^{2y}}$

Now, $\theta = 90° - i$

$\tan\theta = \cot i$

$\frac{dy}{dx} = \cot i$ …(i)

Now, using snell's law

$1 \times \sin 90° = \mu \sin i$

$\sin i = \frac{1}{\mu} = \frac{1}{\sqrt{1+e^{2y}}}$

$\cot i = e^{y}$

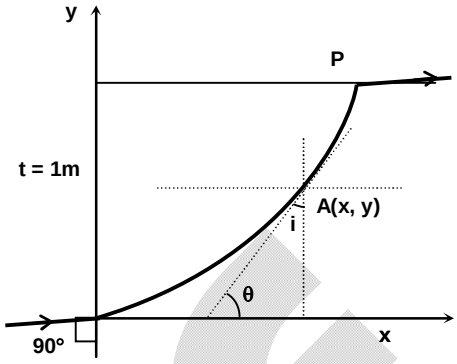
$\frac{dy}{dx} = e^{y}$ (from equation (i))

$\int_0^y e^{-y}dy = \int_0^x dx$

$(1 - e^{-y}) = x \Rightarrow x = (1 - e^{-y})$

Putting y = 1m, $x = (1 - e^{-1}) = \left(\frac{e-1}{e}\right)$

The co-ordinates of the point 'P' at which the light emerges from the slab are $\left(\frac{e-1}{e},1\right)$



3. A, D

Sol. $dN = \left(\frac{N}{b-a}\right)dr$

So, $B_0 = \int_a^b \frac{\mu_0 I}{2r}\frac{N}{b-a}\cdot dr = \frac{\mu_0 IN}{2(b-a)}\ell n\frac{b}{a}$

$M = \int I\pi r^2 \frac{N}{b-a}dr = \frac{I\pi N(b^3-a^3)}{3(b-a)} = \frac{I\pi N}{3}(b^2+a^2+ab)$

4. A

Sol. $V_{rms} = \sqrt{2^2 + \frac{3^2}{2} + \frac{4^2}{2}}$

$\Rightarrow V_{av} = 2$

5. B

Sol. $V = \frac{P}{i} = \frac{P}{ne}$

$V = \frac{1}{6.24\times10^{13}\times1.6\times10^{-19}}$

$V = 10^5$ volts

$\Rightarrow \lambda_{min} = \frac{hc}{eV} = \frac{12431}{V}$ Å ≈ 0.12 Å

6. B

Sol. Potential across each capacitor is 10 V

Charge = 20 μC

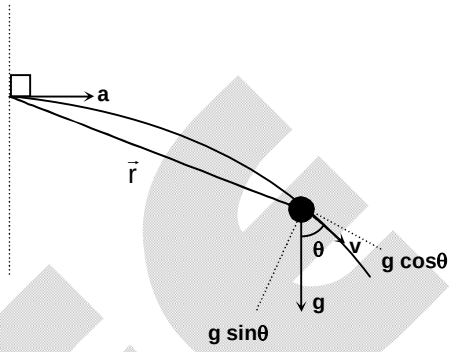
Charge on plate C = 40 μC (both sides included)

7. B

Sol. $W = \int_0^I \mu 2\pi r x dx P = \pi\mu r P \ell^2 = \pi(0.3)(3\times10^{-2})\frac{10^5}{\pi}\times10^{-2} = 9J$

8. A

Sol. θ is always less than 90° (except at projection point)
$\Rightarrow g\cos\theta > 0 \Rightarrow v_{radial}$ always increases
$\Rightarrow g\sin\theta > 0 \Rightarrow v_{ransverse}$ always increases
Also we can see that θ is decreasing with time
So, $a_{transverse} = g\sin\theta$ decreases
$A_{radial} = g\cos\theta$ increases



9. A

Sol. (P) $z = R + (X_L - X_C)i = 3 + (8-4)i = 3 + 4i$

$|z| = \sqrt{(3)^2 + (4)^2} = 5\,\Omega$

$I_{rms} = \frac{\varepsilon_{rms}}{|z|} = \frac{100}{5} = 20\ A$

(Q) $\frac{1}{z} = \frac{1}{X_L i} + \frac{1}{-X_C i} + \frac{1}{R} = \frac{1}{6i} - \frac{1}{3i} + \frac{1}{8} = \left(\frac{1}{3} - \frac{1}{6}\right)i + \frac{1}{8} = \frac{1}{8} + \frac{1}{6}i$

$\Rightarrow z = \frac{48}{(6+8i)}$

$\Rightarrow z = \frac{48(6-8i)}{100}$

$\Rightarrow |z| = \frac{48}{100}\times10$

$\Rightarrow |z| = 4.8\,\Omega$

$I_{rms} = \frac{\varepsilon_{rms}}{|z|} = \frac{120}{4.8} = 25\ A$

(R) $\frac{1}{z'} = \frac{1}{X_L i} + \frac{1}{-X_C i} = \frac{1}{6i} - \frac{1}{3i} = i\left(\frac{1}{3} - \frac{1}{6}\right) = \frac{i}{6}$

$z' = \frac{6}{i} = -6i$

$z = R + z' = 8 - 6i$

$|z| = \sqrt{(8)^2 + (6)^2} = 10\,\Omega$

$I_{rms} = \frac{\varepsilon_{rms}}{|z|} = \frac{50}{10} = 5\ A$

(S) $z_1 = R_1 + X_L i = 8 + 6i \Rightarrow I_{1(rms)} = \frac{\varepsilon_{rms}}{|z_1|} = \frac{100}{10} = 10\ A$

$z_2 = R_2 - X_C i = 3 - 4i \Rightarrow I_{2(rms)} = \frac{\varepsilon_{rms}}{|z_2|} = \frac{100}{5} = 20\ A$

$$\therefore\ I_{rms} = \sqrt{I_{1(rms)}^2 + I_{2(rms)}^2} = \sqrt{(10)^2 + (20)^2} = \sqrt{500} = 10\sqrt{5}\ A$$

10. A

Sol. $A \to B$

$TP^2 = \text{constant}$

$PV^{1/3} = \text{constant}$

$$W_{A\to B} = \frac{P_1V_1 - P_2V_2}{x-1} = \frac{nR(T_1 - T_2)}{x-1} = -\frac{9}{2}nRT_0$$

$$\Delta U_{A\to B} = -3nRT_0$$

$$\Delta Q_{A\to B} = -\frac{15}{2}nRT_0$$

$C \to A$

$TP^4 = \text{constant}$

$PV^{1/5} = \text{constant}$

$$W_{C\to A} = \frac{nR(T_1 - T_2)}{x-1} = \frac{15}{4}nRT_0$$

$$\Delta U_{C\to A} = nR(T_A - T_C) > 0$$

$$\Delta Q_{C\to A} > 0$$

$B \to C$

$\Delta U = 0$

$W_{B\to C} > 0$

$\Delta Q_{B\to C} > 0$

11. C

Sol. (P) frequency of sound receive by the reflector

$$f_r = \frac{v - v_r}{v} f$$

The frequency of reflected sound

$$f' = \frac{v}{v + v_r} f_r = \frac{v - v_r}{v + v_r} f = \frac{330 - 33}{330 + 33} \times 660$$

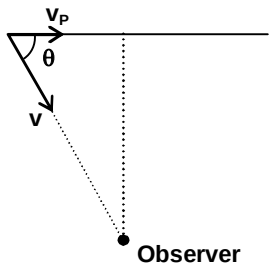
(Q) frequency of sound reaching the hill

$$f_h = \frac{v}{v - v_b} f$$

The frequency of reflected sound

$$f' = \frac{v + v_b}{v} f_h = \frac{v + v_b}{v - v_b} f = \frac{330 + 16.5}{330 - 16.5} \times 627 = 693\ Hz$$

(R) $$f' = \frac{v}{v - v_P \cos\theta} f = \frac{v}{\left(v - v_P \frac{v_P}{v}\right)} f = \frac{v^2}{v^2 - v_P^2} f$$



$$\frac{v^2}{v^2 - \left(\frac{v}{2}\right)^2} f = \frac{4}{3} \times 450 = 600\ Hz$$

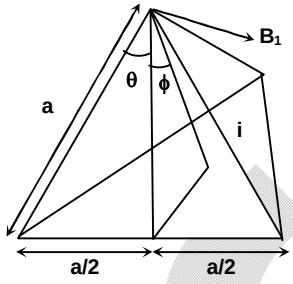
(S) $$f' = \sqrt{\frac{c - v}{c + v}} f = \sqrt{\frac{c - \frac{c}{2}}{c + \frac{c}{2}}} \times 500\sqrt{3} = 500\ GHz$$

SECTION – B

12. 3

Sol. Magnetic field due to one side

$$B_1 = \frac{\mu_0 i}{4\pi\left(a\frac{\sqrt{3}}{2}\right)} 2\sin 30^\circ = \frac{\mu_0 i}{2\sqrt{3}\pi a}$$



Resultant magnetic field

$$B = 3B_1\cos\left(\frac{\pi}{2}-\phi\right) = 3\left(\frac{\mu_0 i}{2\sqrt{3}\pi a}\right)\left(\frac{\frac{a}{2\sqrt{3}}}{\frac{a\sqrt{3}}{2}}\right) = \frac{\mu_0 i}{2\sqrt{3}\pi a}$$

13. 4

Sol. $a_E = \frac{qE_0}{m}$ (along negative z-direction)

Electric field will make z-component of velocity zero. At that time speed of the particle will be minimum and that minimum speed is the other component, i.e. v_0.

This is minimum when,

$v_z = u_z + a_z t$

Or $0 = v_0 - \frac{qE_0}{m}t$

Or $t = \frac{mv_0}{qE_0}$

14. 5

Sol. If f_{eq} is the focal length of combination, we use

u = –12.5 cm

For real image v = mu = +50 cm

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_{eq}}$$

$$\Rightarrow \frac{1}{50} + \frac{1}{12.5} = \frac{1}{f_{eq}}$$

$$\Rightarrow \frac{1}{f_{eq}} = \frac{1}{10}$$

$\Rightarrow f_{eq}$ = 10 cm

Equivalent power of the lens system is

$$P_{eq} = \frac{1}{0.1} = 10D = 2P_{eqcu}$$

Optical power of each lens is

P_{eqch} = 5D

15. 4

Sol. Energy released = Final binding energy – initial binding energy

= 110 × 8.2 + 90 × 8.2 – 200 × 7.4 = 160 MeV

16. 50

Sol. $a = \dfrac{\mu mg}{M + 2m}$

$F = 2(M + m)\dfrac{\mu mg}{(M + 2m)}$

$\Rightarrow F = 50$ N

17. 3

Sol. $T = \dfrac{2\pi}{\omega} = 2\pi\sqrt{\dfrac{\ell}{3g}}$

Chemistry

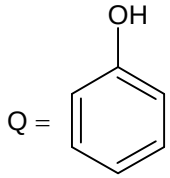
PART – II

SECTION – A

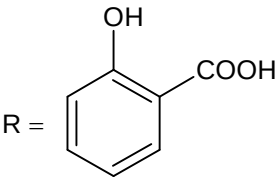
18. A, D
Sol. For solid and liquid $C_P \approx C_V$
Below critical temperature, gases may be liquefied through application of pressure. Thus, $Z < 1$.

19. A, B, C
Sol. A, B and C have alpha hydrogen, so will give aldol. D has no alpha hydrogen.

20. A, B, C, D
Sol.



Q = , $P = CH_3COOH$



R =

21. A
Sol. Conceptual

22. A
Sol. At anode
$2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$
At cathode
$2H_2O + 2e^- \longrightarrow H_2 + 2OH^-$

23. B
Sol. Due to intensive hydration of F^- ion, because of smallest size.

24. D
Sol. Acetal give negative test.

25. A
Sol. Named reaction.

26. C
Sol. Conceptual

27. D
Sol. (P) $P_2O_3 + H_2O \longrightarrow H_3PO_3$
(Q) $\text{Red } P_4 + \text{alkali} \longrightarrow H_4P_2O_6$
(R) $\text{White } P_4 + \text{alkali} \longrightarrow H_3PO_2$
(S) $PCl_3 + H_3PO_3 \longrightarrow H_4P_2O_5$

28. A
Sol. Conceptual

SECTION – B

29. 6
Sol. XeO_6^{4-}, 90° bond angles are 12.
In PCl_3F_2, 90° bond angles are 6.
In ClF_3, 90° bond angles are 0.
$\therefore 12-6-0=6$

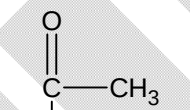
30. 3
Sol. (i), (ii) and (iii)
(i) Average kinetic energy is a function of temperature, not of molar mass.
(ii) At constant Temperature and pressure
$d \propto M$, density of gas P will be four times of gas Q.
(iii) Equal volume of P and Q will have equal moles at constant temperature and pressure, therefore, mole fraction of each component in mixture will be 0.5
Partial pressure = 0.5 × 1 = 0.5 atm.

31. 2
Sol. $XeF_6 \xrightarrow[\text{hydrolysis}]{\text{partial}} XeOF_4 + 2HF$
$XeF_6 \xrightarrow[\text{hydrolysis}]{\text{complete}} XeO_3 + 6HF$

32. 4
Sol. N_2O, N_2O_4, $H_2S_2O_3$ and $H_2S_2O_6$

33. 11
Sol. $A = CH \equiv CH$

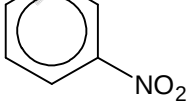
B =

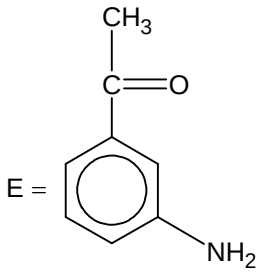


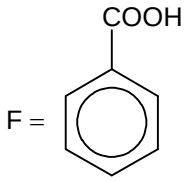
C =



D =







34. 200

Sol. $P_T = \sqrt{P_A^o . P_B^o}$

Mathematics

PART – III

SECTION – A

35. B, C, D

Sol. $P(n)=\left(\frac{1}{2}\right)\left(\frac{2}{3}\right)\left(\frac{3}{4}\right).....\left(\frac{n-2}{n-1}\right)\left(\frac{n-1}{n}\right)\left(\frac{1}{n+1}\right)=\frac{1}{n(n+1)}$

$\therefore \frac{1}{n(n+1)}<\frac{1}{2024} \Rightarrow \frac{2}{n(n+1)}<\frac{1}{1012} \Rightarrow \frac{n(n+1)}{2}>1012$

36. A, D

Sol. $f(x)=-\frac{x^2}{2}-\frac{x}{2}+5$

37. A, B, C

Sol. Each of the 2n + 2 sums in question lies in the (2n + 1) elements {–n, –n +1, ... –1, 0, 1, ... n – 1, n}. Hence, by P..H..P two of these sums have same values

38. C

Sol. ${}^{2024}C_{1012}=2^7\ 3^0\ 5^1\ 7^3$

39. A

Sol. $x_{2025}=x_5$, $x_{2026}=-x_1$, $x_{2027}=-x_2$

$\therefore |267-211-375|=319$

40. B

Sol. $\frac{1}{2}\times|z_2-z_3|\times|az_2+(1-a)z_3-z_1| = \frac{|z_1-z_2||z_2-z_3||z_1-z_3|}{4R}$

41. D

Sol. $\frac{\frac{1-a}{1+a}+\frac{1-b}{1+b}+\frac{1-c}{1+c}+\frac{1-d}{1+d}+\frac{1-e}{1+e}}{5}\geq\frac{1-\frac{1}{5}}{1+\frac{1}{5}}$

42. D

Sol. (P) $\int_0^1\left(f^3(x)-4x\right)^2dx=0$

(Q) $g(x)=\frac{x^5}{5}\int_1^x f(u)du$

(R) $\int_0^1\frac{t^3}{\ln t}dt-\int_0^1\frac{1}{\ln t}dt=\int_0^1\frac{t^4}{t\ln t}dt-\int_0^1\frac{t}{t\ln t}dt=0$

(S) $I_2=\int_0^1\frac{t^2}{e^{t^3}(2-t^3)}dt\ ;\ t^3=z$

$\Rightarrow\frac{1}{3}\int_0^1\frac{dz}{e^z(2-z)}$

43. B

Sol. (P) Angle between line and plane is 45°

$\therefore$ Minimum distance = $\frac{|\sqrt{2}+7-\sqrt{2}-1|}{2}=3$

(Q) $\left[\vec{a}\times\vec{b}\quad\vec{b}\times\vec{c}\quad\vec{c}\times\vec{a}\right]=\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]^2$

For maximum value $\vec{c}$ is perpendicular to $\vec{a}$ and $\vec{b}$ thus $\vec{c}$ is parallel to $\vec{a}\times\vec{b}$

Hence, $\left|(\vec{a}\times\vec{b})\times\vec{c}\right|=0$

(R) $\left[\vec{a}\ \vec{b}\ \vec{c}\right]=2$

(S) The point of intersection with planes

$\left(a\hat{i}-\hat{j}+7\hat{k}\right)\cdot\vec{r}=1$, $a\hat{i}+\hat{j}-2\hat{k}=-1$ and $\hat{k}\cdot\vec{r}=0$ is (0, –1, 0)

The point of intersection $\left(2\hat{i}-\hat{j}+2\hat{k}\right)\cdot\vec{r}=5$, $\hat{k}\cdot\vec{r}=0$ and $\left(a\hat{i}-\hat{j}+7\hat{k}\right)\cdot\vec{r}=1$

$\left(\frac{4}{2-a},\frac{5a-2}{2-a},0\right)$. This point lie on x-axis $\Rightarrow a=\frac{2}{5}$

44. C

Sol. (P) $n(E_2) = {}^9C_7 = 36$

$n(E_1$ of $E_2)$

= {matrix formed by 7 one's and 2 O's and both 3 are at same row of same column}

$= 6\times {}^3C_2 = 18$

$P\left(\frac{E_1}{E_2}\right)=\frac{1}{2}$ $\therefore$ p + q = 3

(Q) $\begin{vmatrix}1 & a & a^2\\ 1 & b & b^2\\ 1 & c & c^2\end{vmatrix}-\frac{1}{abc}\begin{vmatrix}a & a^2 & abc\\ b & b^2 & abc\\ c & c^2 & abc\end{vmatrix}=0$

(R) $-1(1-\cos^2 P)+\cos R(\cos Q\cos P+\cos R)+\cos Q(\text{Cos R cos P + cos Q})$

$=\left(\sum\cos^2 P\right)+2\cos P\cos Q\cos R=1$

(S) $(B - rI) = (n - r)A$

$|(n - r)A((n - r)A - n(n - r)I)| = |(n - r)^2A^2 - n(n - r)^2A|$

$= |n(n - r)^2A - n(n - r)^2A| = 0$

45. D

Sol. (P) Area of $\Delta PTS) \times$ Area $(\Delta PTS') = \frac{(PT)^2\times b^2}{4}$

(Q) $3am - 2am - am^3 = 0$

$m = 0, 1, -1$

$\therefore k - 4 = 4$

(R) $3\left|h-1+\frac{1}{h-1}+2\right| \Rightarrow \Delta_{min}=12$

(S) For $x + 2y - a = 0$ to intersect the circle $x^2 + y^2 = 4$, $\left|\frac{-a}{\sqrt{5}}\right|\le 2$

$\therefore -2\sqrt{5} < a < 2\sqrt{5}$

Also, radical axis $S_1 - S_2 = 0 \Rightarrow 4x + 2y - 5 = 0$

For A, B, C, D to be concyclic radical $x + 2y - a = 0$ and $12x - 6y - 41 = 0$ must be concurrent

$\begin{vmatrix}1 & 2 & -a\\ 12 & -6 & -41\\ 4 & 2 & -5\end{vmatrix}=0 \Rightarrow a = -2$

SECTION – B

46. 8

Sol. $\alpha_n = {}^nC_4 \cdot 2^{n-4}$

$\Rightarrow$ Required sum = $3\left[\frac{2}{{}^4C_4}+\frac{2^2}{2\cdot{}^5C_4}+\frac{2^3}{2^2\cdot{}^6C_4}+.....\right] = 6\left[\frac{1}{{}^4C_4}+\frac{1}{{}^5C_4}+\frac{1}{{}^6C_4}+.....\infty\right]$

$= 6\left[\sum_{r=4}^{\infty}\frac{1}{{}^rC_4}\right] = 6\lfloor 4\sum_{r=4}^{\infty}\frac{1}{r\cdot r-1\cdot r-2\cdot r-3} = 8$

47. 0

Sol. The equation is $x^2 - (a^2 + 1)x + 4 = 0$

$\therefore$ f(1) > 0, f(4) > 0 $\Rightarrow 1 < \frac{a^2+1}{2} < 4$ and $D \geq 0$

$\therefore\ a \in \left(-2, -\sqrt{3}\right] \cup \left[\sqrt{3}, 2\right)$

48. 2

Sol. $t_1 + t_2 + ... + t_n = 465 = \frac{n}{2}(t_1 + t_n)$ ..... (1)

and $(t_1+1)+(t_2+3)+.....+(t_n+(2n-1)) = 690$ ..... (2)

Subtracting equation (2) – (1), we get $\sum_{n=1}^{n}(2n-1) = 225 \Rightarrow n = 15$

$\therefore\ \frac{t_1+t_n}{31} = 2$

49. 0

Sol. Let $x = \alpha$ is a common root

$\alpha^2 + a\alpha + b = 0$ ..... (1)

$\alpha^2 + \alpha + ab = 0$ ..... (2)

$a\alpha^2 + \alpha + b = 0$ ..... (3)

Subtracting equation (1) – (2), we get $(\alpha - 1)(\alpha - b) = 0$

(1) or (3) gives $b^2 + ab + b = 0$ and (3) gives $ab^2 + 2b = 0$

$b \neq 0 \Rightarrow ab = -2$, $ab = -2$, $b^2 + b - 2 = 0$, b = 1 or –2, a = –2 or 1

a = –2, b = 1 and a = 1, b = –2 (not satisfy the condition)

$\therefore$ a + 2b = 0

50. 4

Sol. $z^4 + (5z - 1)^4 \Rightarrow (-1)^{\frac{1}{4}} = \cos\frac{2m\pi+\pi}{4} + i\sin\frac{\pi+2m\pi}{4} \Rightarrow m = 0, 1, 2, 3$

$\therefore\ \sum\frac{1}{\alpha_1\beta_1} = \frac{104}{2} = 52 \Rightarrow f : R \to R$

51. 5

Sol. $f(\sin\pi x) = \begin{cases} 0 & ;\ \sin\pi x < \frac{1}{2} \\ 1 & ;\ \sin\pi x \geq \frac{1}{2} \end{cases} \Rightarrow \int_0^1 f(\sin\pi x)\,dx = \int_0^{\frac{1}{6}} 0 + \int_{\frac{1}{6}}^{\frac{5}{6}} 1 + \int_{\frac{5}{6}}^{1} 0 = \frac{2}{3} \Rightarrow p + q = 5$